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TRACEABILITY & SERIALIZATION MODULE

Automotive Manufacturer Software Platform

End-to-End Traceability • Serialization • Quality Integration • Compliance

IATF 16949 Compliant | Industry 4.0 Ready | OEM Trusted

AGENDA

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01
End-to-End Traceability
Forward & backward traceability across the supply chain

02
Serialization Management
Unique ID generation and serial lifecycle control

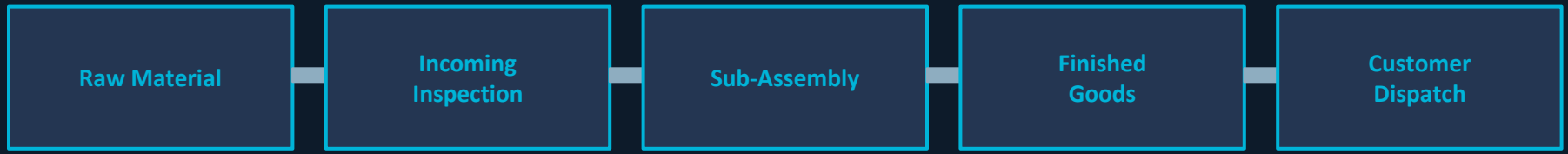
03
Production & Process Traceability
Machine parameters, MES integration & shop-floor data

04
Quality & Compliance
Quality event linkage, recall management & audit readiness

05
Supplier & Customer Traceability
Multi-tier supplier mapping & OEM portal integration

06
Advanced Capabilities & Benefits
Digital twin, AI anomaly detection & business outcomes

01 | END-TO-END TRACEABILITY



← BACKWARD TRACEABILITY | FORWARD TRACEABILITY →

PART & PRODUCT TRACEABILITY

- Serial, batch & lot-level traceability
- One-up / One-down traceability (IATF)
- Finished goods ↔ sub-assembly ↔ component
- Multi-level BOM traceability
- Vehicle VIN / Engine / Chassis traceability

MATERIAL TRACEABILITY

- Raw material heat / batch tracking
- Supplier lot & certificate linkage
- Incoming inspection result linkage
- FIFO / FEFO / LIFO traceability support
- Material genealogy from receipt to shipment

02 | SERIALIZATION MANAGEMENT

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CREATE → PRODUCE → INSPECT → REWORK → SHIP → WARRANTY

SERIAL LIFECYCLE

Auto-Generation Rules

Customer-specific serial formats with date, shift, plant & line encoded serials

Barcode / QR / RFID

Multi-format unique ID support for all scan technologies across the shop floor

Rework & Scrap Handling

Re-serialization on rework with complete history preservation and audit trail

Duplicate Prevention

Real-time duplicate detection with automatic blocking before any downstream use

Serial Blocking

Quarantine and blocking controls with instant notification and release workflow

Lifecycle Tracking

Creation through warranty — complete end-to-end serial state management

03 | PRODUCTION & PROCESS TRACEABILITY

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PROCESS PARAMETER CAPTURE

Torque	Captured per station
Temperature	Real-time monitoring
Pressure	With tolerance alerts
Speed / RPM	Machine-level tracking
Operator ID	Shift & workstation log
Tool / Mold	Die & program version

MES / SHOP-FLOOR INTEGRATION

- Automatic data capture from machines
- Station-by-station serial tracking
- Line clearance verification
- Interlock-based process enforcement
- Real-time production genealogy
- Recipe / program version tracking

04 | QUALITY TRACEABILITY INTEGRATION



QUALITY EVENT LINKAGE

RECALL & CONTAINMENT MANAGEMENT

- Instant affected serial identification
- Recall simulation & impact analysis
- Lot containment & blocking
- Customer notification readiness
- Cost impact estimation

05 | SUPPLIER & CUSTOMER TRACEABILITY

SUPPLIER TRACEABILITY

- Supplier lot to finished part mapping
- Supplier PPAP & certificate linkage
- Supplier quality issue impact tracing
- Multi-tier supplier traceability

CUSTOMER & OEM TRACEABILITY

- Customer dispatch serial mapping
- ASN & e-way bill linkage
- OEM portal integration
- Warranty claim traceability

Full multi-tier traceability — from raw material supplier to end customer

06 | COMPLIANCE, AUDIT READINESS & DASHBOARDS

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IATF 16949

OEM Mandates

Safety & Regulatory

Digital Audit Trail

STANDARDS & COMPLIANCE COVERAGE

DATA RETENTION & SECURITY

- Configurable data retention periods
- Tamper-proof audit logs
- Role-based access control
- Electronic signatures

DASHBOARDS & ANALYTICS

Serial Genealogy View	*Single click*
Traceability Score	*Completeness %*
Recall Readiness KPI	*Live tracking*
Quality Heatmap	*Issue mapping*
Scrap & Rework Trends	*Analytics*

07 | ADVANCED CAPABILITIES (DIFFERENTIATORS)

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Digital Twin

Complete digital twin of product genealogy — every part, process and parameter captured virtually

Blockchain-Ready

Blockchain-ready traceability framework for immutable, tamper-proof record keeping

AI Anomaly Detection

AI-based anomaly detection spots quality deviations before they become defects or recalls

Mobile Scanning

Handheld & tablet scanning for shop-floor operators — real-time barcode, QR & RFID capture

Offline Mode

Offline shop-floor mode ensures uninterrupted operations even during network disruptions

08 | BUSINESS BENEFITS

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Zero Recall Panic

Instant root cause identification — trace any serial in seconds

OEM Trust & Compliance

Confidence across all OEM-specific mandates and IATF 16949

Reduced Warranty Costs

Targeted containment reduces recall scope and associated costs

Faster Audits

Complete digital trail shortens audit cycles and customer approvals

Complete Genealogy

Every product has a full digital history from material to delivery

Industry 4.0 Ready

Digital twin, AI & blockchain-ready foundation for smart factory

09 | TYPICAL USE CASES (DEMO READY)

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01 Engine Assembly

Torque traceability

Every torque fastening captured per station with operator ID, tool, and parameter — fully reconstructable

02 Safety Parts

Airbag / Brake Serialization

Critical safety part serialization with 100% genealogy — zero ambiguity in any recall scenario

03 EV / Battery Pack

Component traceability

Cell-to-pack traceability for EV batteries with full supplier certificate and test result linkage

04 Recall Simulation

Under 60 seconds

Simulate any recall scenario and instantly identify affected lots, serials, customers and cost impact

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READY TO TRANSFORM YOUR TRACEABILITY?

Schedule a live demo and see end-to-end traceability in action.

- ✓ IATF 16949 Compliant
- ✓ OEM Trusted
- ✓ Industry 4.0 Ready
- ✓ Recall in < 60 Seconds